Kahnrad Braxton

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Education

Johns Hopkins University — Baltimore, MD
Ph.D. in Mathematics — *Expected May 2027*

- Awarded **The Kelly Miller Fellowship** for outstanding contributions from underrepresented graduate students to the intellectual life of the department and broader community
- Awarded **The Professor Joel Dean Award for Excellence in Teaching Mathematics** *(2024, 2026)*, recognizing extraordinary performance in undergraduate education, funded by the Joel Dean Foundation since 1957

Johns Hopkins University — Baltimore, MD
M.S. in Systems Engineering — *Dec 2025*

- **Coursework:** Applied Statistics & Data Analysis, Machine Learning, System Dynamics, Monte Carlo Modeling & Simulation, Computing for Applied Mathematics, Operations Research, Introduction to Systems Engineering, Scientific Machine Learning for Modeling, Optimization, and Control of Dynamical Systems

Johns Hopkins University — Baltimore, MD
M.A. in Mathematics — *May 2023*

University of Pennsylvania — Philadelphia, PA
B.A. in Mathematics — *May 2022*

Publications

Braxton, K., Lee, T. K., Zhu, J. *A stable self-shrinking Möbius bundle in* $\mathbb{R}^4$. Submitted for publication, 2026. arXiv:2609.19648.

Research & Industry Experience

Johns Hopkins Applied Physics Laboratory — Laurel, MD
Systems Engineer Intern — *June 2026 - Present*

- Apply Model-Based Systems Engineering (MBSE) methodologies using SysML and Cameo to model the metadata lifecycle of a reliability analysis system, developing activity diagrams that capture system processes, interfaces, and dependencies
- Develop a Python statistical modeling package by translating and extending Bayesian analysis capabilities from R, supporting quantitative reliability analysis and communicating technical progress through weekly presentations to engineers and stakeholders
- Formulated and mathematically derived an analytical method for quantifying the probability that uncertain, correlated alternatives lie on the Pareto frontier, enabling probabilistic evaluation of competing alternatives in system trade studies
- Translate complex mathematical, statistical, and systems engineering problems into structured modeling and analytical approaches that support system evaluation and technical decision-making
- Final Secret security clearance

Sedaro — Arlington, VA
Modeling & Simulation Engineer Intern — *Jan 2026 – May 2026*

- Developed high-fidelity simulation software and analysis tools for complex engineered systems, with emphasis on model validation, quantitative analysis, and customer-facing operational workflows.
- Configured and executed large-scale multi-simulation studies using cloud compute resources to support uncertainty quantification, sensitivity analysis, and optimization.
- Built an LLM assistant integrated with company-developed tools through an MCP server, enabling customers to plan and analyze strategic operational scenarios.
- Collaborated across engineering, product, and customer-facing teams to design new workflows tailored to evolving stakeholder needs and mission requirements.
- Conducted peer code reviews, supported technical demonstrations, and contributed engineering input during contract meetings with external stakeholders.

Johns Hopkins University Baltimore, MD

Engineering Graduate Research Assistant *Aug 2024 – Present*

- Develop and implement Python algorithms for mathematical modeling and high-fidelity simulation of earthquake ground motions, supporting risk assessment and infrastructure resilience
- Optimize synthetic ground motion generation techniques, improving the physical realism and statistical robustness of seismic hazard simulations
- Validate simulation outputs against historical earthquake data, enhancing model accuracy and ensuring alignment with observed ground motion characteristics
- Design and execute hurricane hazard modeling pipelines, generating large-scale synthetic storm catalogs for scenario-based risk analysis
- Perform statistical regression and sensitivity analysis on hurricane parameters to improve storm modeling accuracy and inform model calibration strategies

Outlier / Snorkel AI Remote

AI Trainer *Jan 2024 – Dec 2025*

- Designed and refined AI models for solving mathematical problems by developing prompts, structuring queries, and constructing rigorous, step-by-step solution paths
- Evaluated AI-generated mathematical content for accuracy, logical consistency, and contextual relevance using advanced subject-matter expertise
- Collaborated with technical and content teams to enhance model interpretability, performance, and the clarity of user-facing mathematical explanations
- Supported AI training and evaluation pipelines through algorithmic reasoning, structured problem-solving, and precise technical writing

Johns Hopkins University Baltimore, MD

Graduate Teaching Assistant *Aug 2023 – Present*

- Instruct courses including Single-Variable Calculus, Multivariable Calculus, Linear Algebra, System Dynamics, Data Science, Real Analysis, Dynamical Systems, and Proof-Based Linear Algebra, engaging over 500 students across multiple sections
- Lead curriculum development and facilitate collaborative, think tank-style sessions to foster conceptual understanding and peer-driven inquiry
- Communicate advanced mathematical and computational ideas with clarity, tailoring explanations to diverse technical backgrounds and learning styles
- Provide consistent, accessible support through office hours and mentorship, promoting an inclusive and supportive learning environment

Skills & Competencies

INCOSE Member

Technical Tools & Languages: Proficient in Python (pandas, scikit-learn, NumPy, PyTorch, TensorFlow), R, MATLAB, and Java; experience with Gitlab, Cameo, SysML (basic), Matplotlib, and seaborn

Core Competencies: Mathematical modeling and quantitative analysis; statistical learning and inference; risk modeling and decision optimization; MBSE; simulation and stochastic processes; machine learning and predictive analytics; large-scale data analysis and visualization; algorithm design, development, and testing

Analytical Strengths: Rapid problem-solving under pressure; strong critical thinking and abstraction skills; interdisciplinary insight at the intersection of mathematics and engineering

Project & Systems Engineering: Complex systems modeling and implementation; technical writing and scientific communication; agile collaboration and team-based innovation; research integrity and cross-functional communication